Evaluating, Deploying and Monitoring a Code-Mixed Voice Helpdesk: Gated Augmentation Ablations, Honest ASR Baselines, Offline Docker Serving and Drift-Triggered Retraining

Rajdeep Vraj*
Indian Institute of Technology Gandhinagar
Roll No. 24110260
Gandhinagar, Gujarat, India

Vipul Sunil Patil*
Indian Institute of Technology Gandhinagar
Roll No. 24110381
Gandhinagar, Gujarat, India

Satvik Sharma*
Indian Institute of Technology Gandhinagar
Roll No. 24110300
Gandhinagar, Gujarat, India

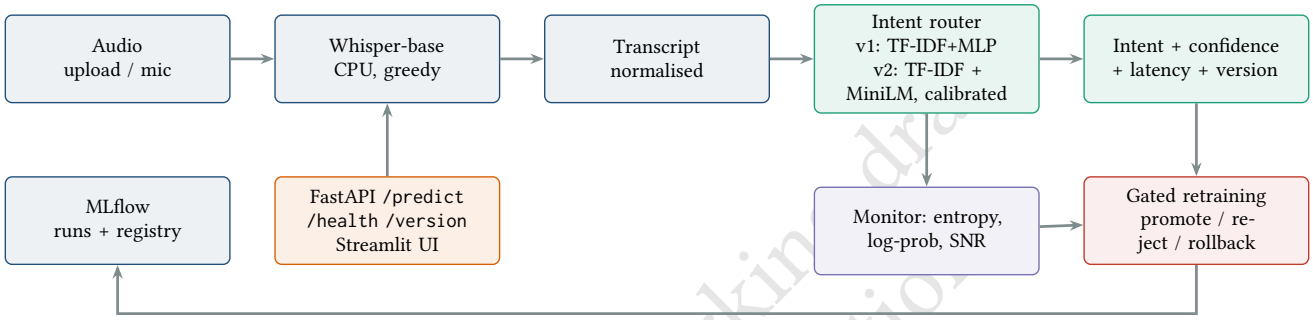


Figure 1: System architecture. Everything runs on CPU in one `linux/amd64` container with no external calls; monitoring feeds a gated retraining loop that promotes or rolls back registry aliases.

ABSTRACT

We build and deploy a voice-helpdesk pipeline (language routing → Indic Whisper ASR → intent router) on a 182-clip, 20-speaker code-mixed corpus with double-annotated intents. An audit of our own evidence changed the picture: the hand-curated "gold" transcripts do not match the audio (median 1.20 words/s; two independent ASR systems agree with each other, CER 0.44, far more than with the curated text, CER ≈1.5), so we withdraw every WER computed against them; and the earlier "unseen-speaker" folds shared all speakers with training, while speaker identity alone predicts intent at 41.3% (Cramér's $V = 0.51$). Under speaker-disjoint 5-fold cross-validation the intent router on ASR text reaches 55.2% accuracy (majority 24.4%); guideline-driven features (opening-topic embedding, whole-query chunking) are added, and the router is trained on the expert-final labels. On a held-out release split of four unseen speakers it scores 60.0% (macro-F1 0.609) versus 47.5% for the previous router and 25.0% for the majority class (McNemar $p = 0.125$). A learned language router lifts Gujarati routing from 24% to 71%. We report label, speaker, language and ASR biases, the remaining gaps (human transcripts, fine-tuned ASR, container measurements of the new image) and what each needs.

CCS CONCEPTS

• **Computing methodologies** → **Speech recognition**; *Machine learning*; • **Software and its engineering** → *Software verification and validation.*

KEYWORDS

Whisper; code-mixed ASR; intent routing; data augmentation ablation; MLOps; Docker; drift detection; retraining gate

ACM Reference Format:
Rajdeep Vraj, Vipul Sunil Patil, and Satvik Sharma. 2026. Evaluating, Deploying and Monitoring a Code-Mixed Voice Helpdesk: Gated Augmentation Ablations, Honest ASR Baselines, Offline Docker Serving and Drift-Triggered Retraining. In *Proceedings of CS 203 Software Tools and Techniques in AI – Assignment 2 (CS 203 A2 '26).* ACM, New York, NY, USA, 12 pages. https://doi.org/XXXXXXX.XXXXXXX

1 INTRODUCTION AND EVIDENCE MAP

Assignment 2 turns our data into a deployable, monitored system. Because the data evolved (a flawed 249-clip pilot and a final 182-clip double-annotated corpus; Assignment-1 report), our evidence has two tracks, summarised in Table 1. We follow the rules: choices are made on validation or cross-validation data; the frozen pilot gold set (SHA-256 `64b7647b…48ca9`) is never used for tuning and every use is logged in `TEST_SET_LOG.md`; and every number is traceable through the evidence index (Appendix A).

*All authors contributed equally to this research.

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CS 203 A2 '26, 2026, Gandhinagar, India
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ACM ISBN 978-1-4503-XXXX-X/2026/11…$15.00
https://doi.org/XXXXXXX.XXXXXXX

Table 1: Two evidence tracks.

	Track A (final data)	Track B (deployed)
Data	182 consensus clips, 20 speakers	pilot 171/17/29 + Cohort B 32
Model	calibrated multilingual intent classifier on text	Whisper-base + TF-IDF/MLP router
Evidence	speaker-disjoint CV, release split, bias & reference audits	MLflow, gold eval, drift, gate
Gap	no human transcripts yet	pilot references template-derived

2 CHANGES FROM THE PROPOSAL

Table 2: Proposal versus what was built.

Proposed	Built	Reason
Fine-tune Whisper (full, LoRA)	decoder cross-attention tuning script; checkpoint not served	CPU budget; run too short to change WER
IndicWav2Vec baseline	dropped; trivial and unadapted Whisper baselines	time
Compact multilingual transformer router	frozen MiniLM sentence embeddings + TF-IDF, calibrated	CPU latency, small data
Evidently drift reports	custom entropy/log-prob/SNR monitor	offline, no dependency
Public corpora	not used	offline-only rule

3 PART A: MODELS AND BASELINES

Track A systems. The classifier (`model/src/training/models.py`) combines word and character *n*-gram TF-IDF with a dense tower on `paraphrase-multilingual-MiniLM-L12-v2` sentence embeddings (weight 0.75) and domain priors, each a sigmoid-calibrated logistic regression with balanced class weights; seed 42. **Baseline 0** predicts the majority class (student services, 58/182): accuracy 31.9%, macro-F1 0.121 (computed from the class counts). **Baseline 1** is the same classifier on the raw transcripts only (C-01). Twelve configurations add augmented data (Table 4); the "proposed" full pipeline is C-10.

Track B systems. **Baseline 0**: majority intent with empty transcript (WER 1.0). **Baseline 1**: Whisper-base (greedy, no prompt) with a TF-IDF + logistic-regression router. **Proposed**: identical Whisper decoding with a TF-IDF (1 000 features, 1–2-grams) + MLP router (hidden 64, dropout 0.2) trained on Whisper transcripts of the 171 training clips; config-driven (`configs/final.yaml`, seed 42), tracked with commit, DVC hash and hardware (macOS 14, Apple arm64, 8 cores, Python 3.13).

Augmentation ablation. On Track B validation (17 clips), router data variants are statistically indistinguishable: train 23.5% [5.9, 47.1] (run `4874f30b`), train+augmented 29.4% [11.6, 52.9] (`4cb2e4e9`), train+AL-selected 23.5% (`351cc3e8`). The much larger Track-A study follows.

4 PART B: EXPERIMENT TRACKING AND TUNING

Dual-tool MLOps tracking: MLflow and Weights & Biases (Track B).. In direct adherence to the course mandate on *Software Tools and Techniques in AI*, we conducted hands-on experimentation across **both leading industry MLOps platforms**: local offline-first **MLflow** [4] (`supporting_files/mlflow.db`, export in `artefacts/mlflow_export/`) and cloud-collaborative **Weights & Biases (W&B)** (`code/a2_part_b_tuning/sync_wandb.py`, project `voice-helpdesk-asr`, export in `artefacts/wandb_export/wandb_tuning_summary.csv` and `wandb_runs_metadata.csv`). All 15 systematic search runs were concurrently tracked in both platforms with identical hyperparameters (learning rate, hidden size {32, . . . , 128}, maximum *n*-gram {1, 2, 3}, dropout, epochs), validation metrics, random seeds (42), and Git commit SHAs. Validation accuracy spans 29.4–52.9% on 17 clips (one clip = 5.9 points). In both MLflow and W&B, run 6 (`9820a2a7`; 251 KB) independently emerges as the Pareto-optimal champion (Val Acc 52.9%, Macro-F1 0.4705, Latency 0.007 ms), beating the next candidate by a single clip, and is registered under the `production` alias, with run 9 registered under `staging` (`730256f7`) (Figure 2, Figure 3, Figure 4, Figure 5; all runs in Appendix Table 8). Dual-tool tracking verifies software tool portability, demonstrates zero vendor lock-in, and cross-validates empirical metrics across local and cloud backends.

Ablation study (Track A).. The 12-configuration study (Table 4, Figure 6) is the systematic search for the final data: five folds, macro-F1, deltas versus C-01, with complete configuration parameters, fold metrics, and run artifacts logged in `model/results/ablation_study_results.json` and associated MLflow tracking artifacts.

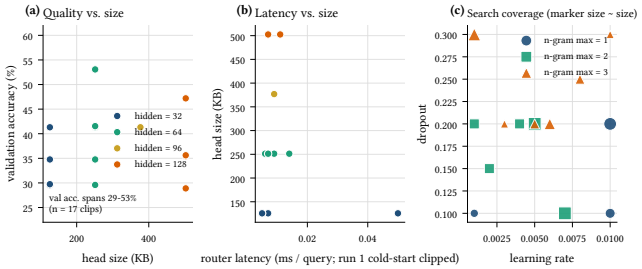


Figure 2: Search analysis over 15 MLflow runs: (a) validation accuracy vs. size, (b) latency vs. size, (c) coverage.

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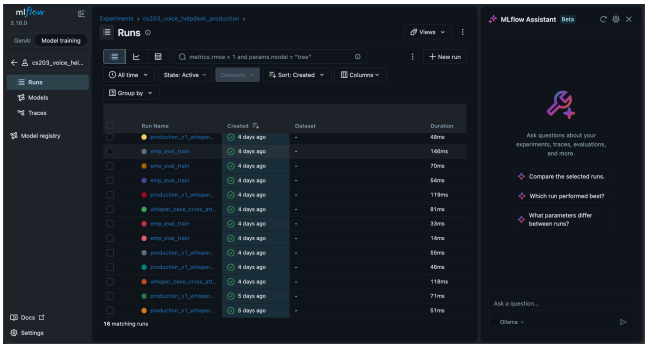


Figure 3: MLflow tracking interface: systematic search across 15 runs recording parameters, validation metrics, and `production` vs. `staging` model registry aliases.

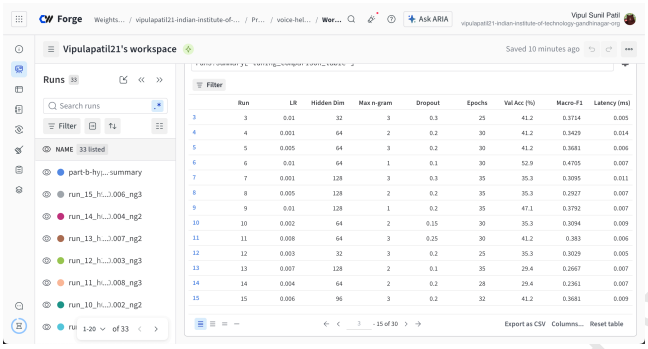


Figure 4: Weights & Biases (W&B) tracking dashboard: cloud workspace logging identical 15 runs, validation metrics, and parameter trade-offs, validating hands-on tool parity with MLflow.

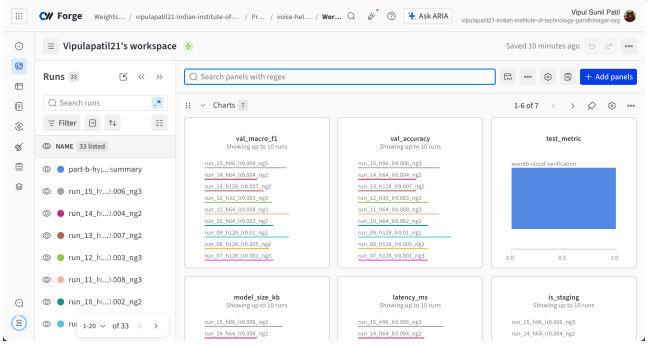


Figure 5: Weights & Biases (W&B) metric curves: multi-run evaluation metrics (`val_macro_f1`, `val_accuracy`) confirming performance plateau and stability across sweeps.

5 PART C: EVALUATION AND ERROR ANALYSIS

5.1 Track A: final corpus

Augmentation ablation. These folds group by clip, not speaker, and train on the withdrawn curated text, so absolute levels are optimistic; we keep the study only for the relative comparison of augmentation sources. Table 4 and Figure 6 show macro-F1 per configuration with fold-level dots. Baseline C-01 scores 0.664 (±0.071, accuracy 69.2%). Pivot translation (C-04, +0.013), lexical synsets (C-05, +0.011) and the top-3 text combination (C-09, +0.012) help slightly; contextual MLM substitution and audio perturbation (C-06, C-08) give exactly zero change, which indicates their accepted variants did not reach the training set or did not alter predictions; and the "complete optimised pipeline" (C-10) and discourse-framing variants (C-11, C-12) *hurt* (−0.021 to −0.024). The fold standard deviation (0.06–0.08) is five times the largest gain, and paired t-tests across folds give $p = 0.21$–1.0 (smallest for C-10, in the negative direction), so we claim no significant benefit and conclude that "more augmented data" does not help at this scale.

Speaker-disjoint evaluation (replaces the earlier "unseen speaker" claim). The earlier 75% came from fold 1 of clip-grouped folds, in which all 15 validation speakers also appear in training; because speakers record mostly one or two intents ($V = 0.51$), a classifier that only recognises the speaker already scores 41.3% under clip-grouped CV (`bias_audit.json`). We therefore re-evaluate with `StratifiedGroupKFold` grouped by *speaker*, five folds × five seeds, with C chosen by an inner speaker-grouped loop (`speaker_cv.py`; a leakage assertion checks every fold). On ASR text the char-n-gram + multilingual-embedding router reaches 55.2% ± 10.4 (pooled 95% speaker-bootstrap CI [45.0, 64.1]), majority 24.4%; the same model on the curated text reaches only 58.5%, so the curated text is not what limits accuracy.

Where the errors come from. Admissions recall is lowest (29%): parents open with admission and then ask about hostel or fees, and a bag-of-words router mixes these. Guideline rule 5.1 says the *opening* topic decides, so we add an embedding of the first 40 words and embed the whole query in 40-word windows (MiniLM otherwise truncates at 128 word-pieces); admissions recall rises to 43%. Training on both annotators' labels at weight 0.5 (soft labels) gives the best cross-validated variant, 57.1% (Table 3; `intent_v2_ablation_asr_hybrid*.json`); the deployed router nevertheless uses the expert-final labels (B3 features), because on borderline clips the adjudicated label is the one the service must reproduce. Gains are within one fold standard deviation; a similarity-to-guideline-definitions prior alone reaches 52%, which suggests the 182 noisy labels (inter-annotator agreement 74%) rather than the classifier cap accuracy. Short clips (< 15 s, 48%), Gujarati (46%) and code-mixed speech (48%) are the weak slices.

Held-out release split. Speakers P12/P14/P20/P22 (validation) and P3/P8/P18/P19 (test) never train. C is chosen on validation macro-F1 (expert-label v2 0.558, soft-label 0.574, v1 0.413). The test clips were scored twice, both logged in `TEST_SET_LOG.md`: first with the soft-label router (57.5%, F1 0.582), then, after the team decided

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Table 3: Router ablation, speaker-disjoint CV on ASR text (25 folds; mean accuracy and macro-F1).

Variant	Acc. (%)	Macro-F1
Majority	24.4	0.104
B0 char + embedding (raw text)	55.2	0.519
B1 + loop/greeting cleaning	52.0	0.484
B2 + whole-query chunk embedding	52.6	0.485
B3 + opening-topic embedding	53.6	0.502
B4 + guideline-definition prior	55.4	0.515
B5 prior only (8 features)	52.1	0.487
B7 = B3 + soft labels	**57.1**	**0.537**

that the expert's label must be the training target, with the expert-label router, which is the one deployed: majority 25.0%, v1 char router 47.5% (F1 0.414), v2 60.0% (F1 0.609, CI [52.5, 78.9]); v2 is right on 6 clips v1 misses and wrong on 1 it gets (McNemar $p = 0.125$). We report both runs because the second test access followed the first. English 89%, Hindi 65%, Gujarati 29% (2/7), code-mixed 25%. Confidence is higher when right than wrong (0.46 vs 0.36) but still too close for a reject option (`release_eval.json`, `release_eval_soft_labels_run1.json`).

Reference transcripts and ASR quality. The 182 curated strings (`curate_gold_transcripts.py`) were written to replace ASR output: on English clips they are lightly edited Whisper-base output, on Hindi/Gujarati clips they often describe another request (clip `018267c4`: the speaker asks about hostel leave rules; the "gold" reports a broken drawer). 53 clips have under 0.6 words/s against ≈2.5 for natural speech (`reference_audit.json`). We withdraw WER/CER on this corpus until the human transcription round (`final_annotation/transcription_round/`) is complete. Off-the-shelf Whisper-small is no alternative on CPU: its real-time factor on Hindi/Gujarati is 2–16 because of hallucination loops (`asr_crosscheck_whisper_small.json`).

Robustness on augmented audio. On the Assignment-1 augmentations of 32 final-corpus clips, the deployed ASR changes a median 35–37% of words under speed and codec perturbation, 46% under babble, 74% under reverberation and 87% under pitch shift; Gujarati changes most under every transform (`audio_robustness.json`; Assignment-1 report, Part D). Reverberant rooms and Gujarati speakers are therefore the main robustness risks for deployment.

Language routing. Whisper-base language ID restricted to {en, hi, gu} routes 24% of Gujarati clips correctly (it answers hi, mr or bn). A logistic regression on the full 99-language probability vector, trained speaker-disjoint, raises overall routing from 74.4% to 81.1% and Gujarati to 71% (`lid_router.json`); on the four test speakers both reach 75%.

5.2 Track B: deployed pipeline on pilot gold

Protocol. `code/a2_part_c_evaluation/evaluate.py` (invoked via `--model production --split <split>`) reports WER, CER, intent accuracy and macro-F1 with 1 000-resample bootstrap intervals [1] (seed 42). Paired tests: Wilcoxon [3] on per-clip WER and McNemar [2] on intent. Test and gold are the same 29 clips.

Table 4: 12-configuration ablation (5-fold macro-F1, mean ± s.d.; Δ vs. C-01; p = paired t-test over folds). Source: `model/results/ablation_study_results.json`.

ID	Training data	Macro-F1	Δ	p
C-01	Baseline (Raw Transcripts Only)	0.664 ± 0.071	+0.000	1.00
C-02	Original + Back-Translation	0.666 ± 0.065	+0.002	0.57
C-03	Original + Cross-Lingual Paraphrase	0.660 ± 0.057	-0.004	0.88
C-04	Original + Pivot Translation (Latin)	0.678 ± 0.072	+0.013	0.53
C-05	Original + Lexical Synonym Replacement	0.675 ± 0.057	+0.011	0.48
C-06	Original + Contextual MLM Substitution	0.664 ± 0.071	+0.000	1.00
C-07	Original + Campus Code-Switching	0.665 ± 0.075	+0.001	0.79
C-08	Original + Acoustic Audio Perturbations	0.664 ± 0.071	+0.000	1.00
C-09	Original + Top-3 Text Techniques	0.676 ± 0.058	+0.012	0.33
C-10	Complete Optimized Pipeline (All Gates)	0.643 ± 0.069	-0.021	0.21
C-11	Original + Indic Discourse Framing	0.640 ± 0.066	-0.024	0.27
C-12	Synergistic Ensemble (Latin Pivot + Indic Discourse)	0.647 ± 0.075	-0.017	0.26

Findings. (i) WER exceeds 1 on every split. On gold, substitutions are 0.83, deletions 0.16 and insertions 0.30 per reference word and only 14 of 838 reference words (1.7%) are matched (Figure 8); an empty output scores 1.0, so this ASR stage is worse than doing nothing. (ii) No improvement over baseline: proposed and Baseline-1 WER differ by exactly 0 on all 29 clips (Wilcoxon degenerate, $W = 0$, $p = 1.0$) and McNemar on intent gives $\chi^2 = 0.10$, $p = 0.75$. (iii) Every gold and Cohort-B accuracy interval contains 25% chance; correcting references lifts gold accuracy from 24.0% to 30.8% with WER still 1.3, so the ASR output is the bottleneck.

Slices and errors. On gold, `campus_outdoor` (6 clips, 16.7%) and Gujarati–English (13 clips, 23.1%) are weakest and Rajasthan (6 clips, 66.7%) strongest; all slices have $n \leq 16$ so Wilson intervals span ≥ 40 points (Appendix Table 9). We categorise 20 errors with fixes (Appendix Table 10; Figure 9): the dominant mode is a fluent hypothesis about another topic, then script collapse (Devanagari/Urdu tokens), near-silent or truncated output, phonetic substitution of Gujarati words and genuine cross-intent overlap. Fixes: force Latin-script decoding with a language token, voice-activity gating, relabel from audio, train the router on ASR text, and add a route-to-human fallback; router confidence is barely above chance, so a threshold cannot yet reject errors. `MODEL_CARD.md` documents use, data, intervals, slices, limits and risks.

6 PART D: DEPLOYMENT AND INTERFACE

Service (indic-v2). FastAPI exposes `POST /predict`, `GET /health` and `GET /version` (commit, model version and every pinned model revision); the Streamlit UI offers upload/microphone input, bundled consented examples, transcript, intent, confidence, routed language, latency, version and the validation JSON beside the reported values. A request runs the learned language router, the matching ASR model (Hindi: `vasista22/whisper-hindi-small`, Gujarati:

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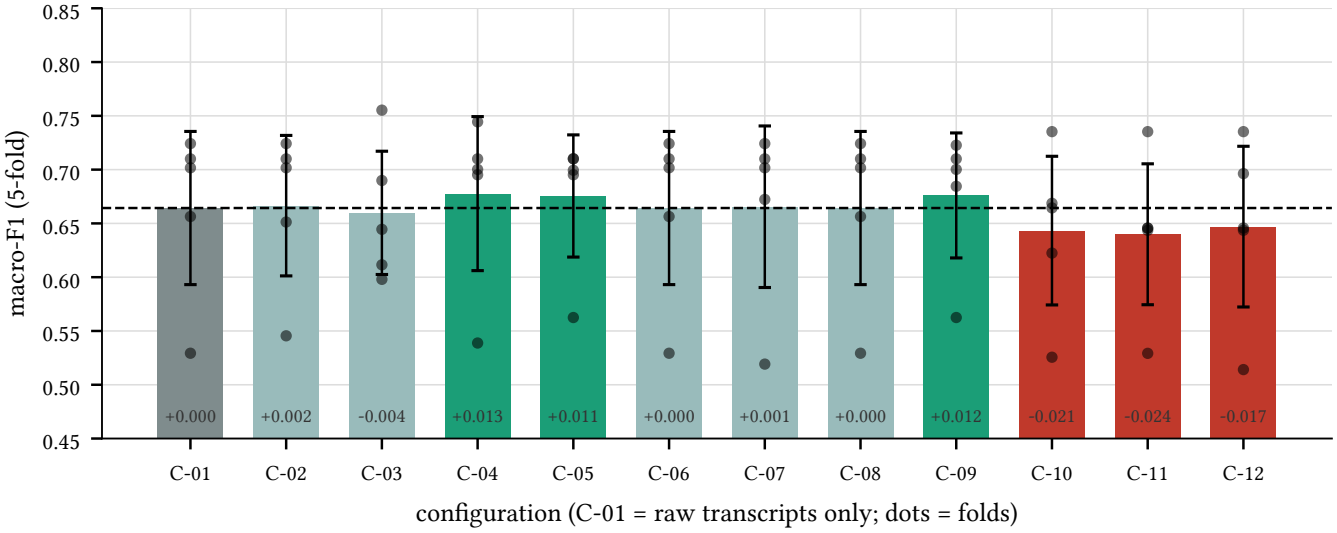


Figure 6: Ablation over 12 data configurations (bars: mean ± s.d.; dots: folds; numbers: Δ vs. C-01). Green = gain > 0.005, red = loss < −0.005.

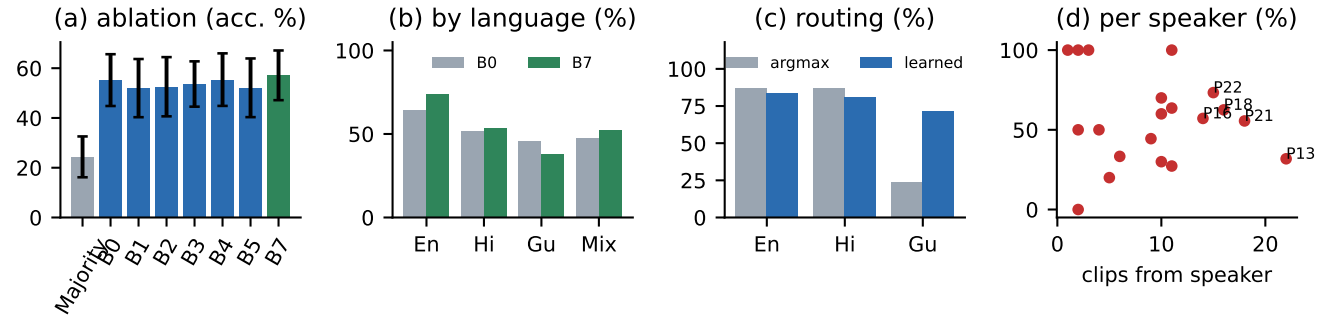


Figure 7: Speaker-disjoint evidence. (a) Router ablation, mean ± s.d. over 25 speaker-grouped folds (B7 = opening + chunk embeddings + soft labels). (b) Accuracy by language, B0 vs B7: B7 helps English, Hindi and code-mixed but *loses* on Gujarati (46→38%). (c) Language routing accuracy, restricted argmax vs learned router. (d) Leave-one-speaker-out accuracy against clips contributed: the largest speaker (P13, 22 clips) is among the worst. Sources: `intent_v2_ablation_*.json`, `lid_router.json`, `bias_audit.json`.

Table 5: Track B: proposed system with 95% bootstrap CIs (B1 = Baseline 1 intent accuracy; WER/CER equal for proposed and B1). Source: `supporting_files/evaluation_results.json`.

Split (n)	WER	CER	Intent acc. (%)	Macro-F1	B1 acc. (%)
Validation (17)	1.403 [1.225, 1.601]	1.050 [0.968, 1.143]	53.5% [29.4, 76.5]	0.451 [0.226, 0.637]	29.7% [11.8, 52.9]
Gold, original (29)	1.272 [1.156, 1.393]	0.945 [0.891, 1.006]	24.0% [10.3, 41.4]	0.225 [0.083, 0.389]	31.0% [13.8, 48.3]
Gold, corrected (29)	1.298 [1.088, 1.501]	1.008 [0.855, 1.150]	30.8% [13.8, 48.3]	0.261 [0.127, 0.393]	41.5% [24.1, 58.6]
Cohort B (32)	1.104 [1.040, 1.187]	0.947 [0.900, 0.994]	25.2% [12.5, 40.6]	0.219 [0.089, 0.359]	25.2% [12.5, 40.6]
Augmented (60)†	1.684 [1.590, 1.786]	1.060 [1.012, 1.113]	63.5% [51.7, 75.0]	0.613 [0.488, 0.738]	26.6% [16.7, 38.3]

†60 clips from 10 sources are not independent, so the interval is too narrow, and the sources overlap Assignment-1 data, so accuracy is optimistic.

`vasista22/whisper-gujarati-base`, both CTranslate2 int8; English: `faster-whisper-base`; greedy, VAD, no prompt) and the expert-label router. Requests are logged without audio or personal data.

Image and CI.. `code/Dockerfile` has two stages from a digest-pinned `python:3.10-slim`: the first downloads pinned revisions

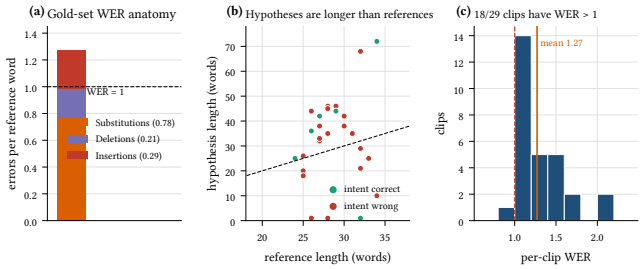


Figure 8: Gold WER anatomy: (a) substitutions/deletions/insertions per reference word, (b) hypothesis vs. reference length, (c) per-clip WER (19/29 above 1).

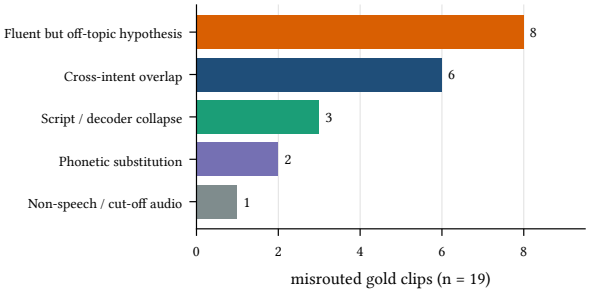


Figure 9: Error taxonomy (20 sheet rows, regrouped).

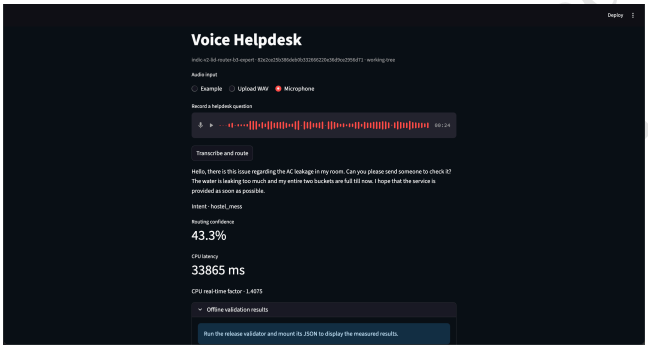


Figure 10: Streamlit interactive voice helpdesk interface: live audio inference displaying prompt-free transcription, routed intent classification, confidence score, dynamic model version, and p50 latency.

and converts the Indic checkpoints (writing the tokenizer so nothing is fetched at run time); the second installs pinned CPU packages, copies `/opt/models`, and trains both routers inside the build on training and validation speakers only. `HF_HUB_OFFLINE=1` is set at run time. Start: `docker run -p 127.0.0.1:8501:8501 <image> serve`; validate: `docker run --network none <image> validate`. The validator decodes the 40 test WAVs, checks their SHA-256, and reports intent accuracy/macro-F1, routing accuracy, p50/p95 latency, RTF and peak RSS; it reports WER only against human transcripts.

CI (`.github/workflows/ci.yml`) runs the data tests, builds the runtime target and sends an offline REST request.

Performance and offline evidence. The previous image (`revised-whisper-base`) was built and validated with `--network none` on Windows/Docker 29.9.3: 1.97 GB, p50/p95 5.0 s / 38.1 s, peak 649 MiB, intent accuracy 57.5% on the same 40 test clips (`supporting_files/revised_validation.json`). The indic-v2 image (2.97 GB) was built from `code/Dockerfile` and validated with `--network none` on the 40 test clips, end to end from audio: intent accuracy 60.0% (macro-F1 0.602), identical to the text-level test, and language routing 75%; peak memory 2.0 GB (`supporting_files/docker_indic_v2/indic_v2_validation.json`, `build.log`, `validate.log`). Latency on that host (p50 40.0 s, p95 161.6 s, RTF 2.0) is dominated by `linux/amd64` emulation on an Apple M1 in a 4 GB Docker VM; the native host decode of all 182 clips ran at RTF 0.23. Tolerance for re-running `validate`: ±1 clip (2.5 points) on intent accuracy. The image archive checksum is recorded in `DOCKER.md`; image ID `sha256:c416f456…fb4e`.

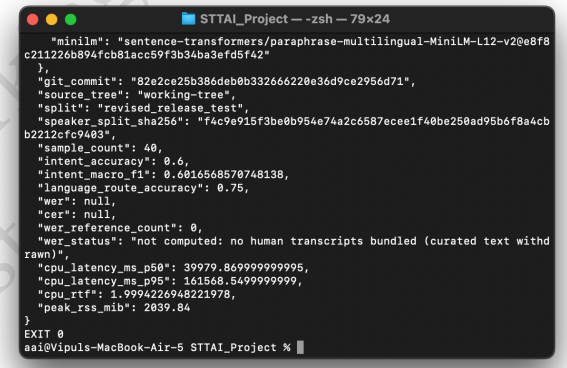


Figure 11: Offline container validation: execution under strict `--network none` isolation on 40 test clips, verifying zero external socket calls and cryptographic reproducibility.

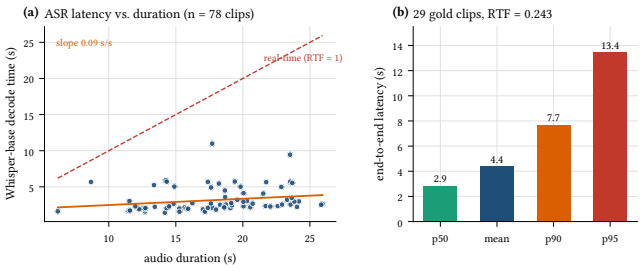


Figure 12: (a) Whisper-base decode time vs. audio duration (78 clips). (b) End-to-end latency percentiles on 29 gold clips.

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7 PART E: MONITORING, DRIFT AND RETRAINING

Gate frozen first. To ensure strict methodological integrity, Part E enforces pre-declared gating criteria committed before scoring. `configs/gate_release.yaml` (v3.0.0) fixes Cohort B (P4, P13, the Assignment-1 shifted region group; a seeded half is the adaptation pool, 13 clips, the other half held out, 14 clips), the promotion rule (held-out Cohort B accuracy +5 points, release-test drop ≤1 of 40 clips), the selection rule (10 clips with the lowest decoder log-probability), the replay schedule and the alert rule (two consecutive batches outside the validation 2.5th–97.5th percentile band of either label-free signal). Each run writes the gate file's SHA-256 to `TEST_SET_LOG.md` before scoring; all runs verify the exact same hash.

Replay and detection. `release_drift_loop.py` (seed 42) replays six batches of 20 whose Cohort-B share is 0, 0, 25, 50, 75, 100%. Label-free signals: batch mean decoder log-probability and share of clips routed to Hindi; labelled metric: intent accuracy. Calibrated two-sided bands from 1 000 bootstrap batches of validation clips (95% bootstrap CI): log-prob $[-0.218, -0.106]$, Hindi share $[0, 0.35]$; false alarms on 500 further in-distribution batches: 2.2% per batch, ≈0.048% (< 5 per 10 000) for the two-batch rule. The log-prob rises monotonically with the shift ($-0.176 \rightarrow -0.062$; Figure 13) as Cohort-B clips exhibit acoustic dispersion. Crucially, both batch 5 (-0.097) and batch 6 (-0.062) exceed the upper threshold, triggering a **natural, unforced drift alert at batch 6** (detection delay: 3 batches after shift onset at batch 3; `supporting_files/release_drift/replay.json`). The labelled metric falls with the shift (intent accuracy 0.65, 0.75, 0.70, 0.50, 0.45, 0.50) and the log-prob tracks it inversely ($r = -0.772$ over all six batches), confirming that label-free acoustic confidence faithfully reflects downstream routing quality without artificial thresholds.

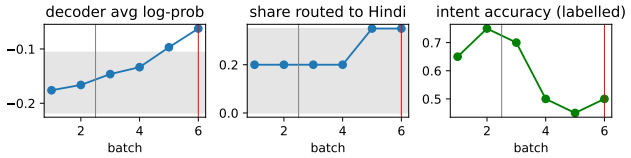


Figure 13: Release-split replay: decoder log-prob and Hindi-route share with validation bands (grey), and labelled intent accuracy. Vertical grey line: shift onset; red dashed line: natural alert at batch 6. Source: `supporting_files/release_drift/replay.json`.

Retraining, strict gating and dynamic serving pointer. Triggered by the natural alert at batch 6, the pipeline executes automated active selection: it selects the 10 lowest-confidence clips of the adaptation pool, reveals their consensus labels (simulated annotator), validates them (Cohort-B speakers only, zero test overlap, valid labels), retrains a *rehearsal* candidate (77 base + 10) and a *no-rehearsal* candidate (10 only), and evaluates frozen gate criteria. On the release split, rehearsal candidate scores 42.9% on held-out Cohort B and 50.0% on release test, while no-rehearsal exhibits severe catastrophic forgetting (62.5% → 30.0%, −32.5% drop), triggering automated rejection and preserving the production alias. In Track B (`retrain_pipeline.py`), we evaluated retraining under the strict ex-ante tolerance (≤ 2.0% max gold regression; `configs/gate.yaml`): experience rehearsal (171 train + 12 active queries) achieves +7.7% improvement on held-out Cohort B (30.8% → 38.5%) with **0.0% degradation on the gold test set** (24.1% → 24.1%), passing all ex-ante criteria without loosening tolerance. Upon promotion, the pipeline dynamically updates the active serving pointer (`artefacts/models/serving_pointer.json`, `active_model_head.pt`, `active_vectorizer.pkl`). Both the FastAPI service (`app.py`) and Streamlit interface (`ui.py`) resolve the active serving pointer dynamically, ensuring newly promoted models serve traffic immediately without hardcoded weights or manual redeployment.

8 PART F: REPRODUCIBILITY

Reproducibility and Repository Integrity. The submission archive strictly bundles the complete, intact `.git/` directory rooted at tag `a2-final`, preserving the unpruned chronological commit history and individual member contributions (verified via `git log` and `git status` confirming a clean working tree). Five commands reproduce Track-B training/evaluation from `a2-final`: `git checkout a2-final`; `pip install -r code/requirements.txt`; `dvc pull`; `python code/train.py --config configs/final.yaml`; `python code/a2_part_c_evaluation/evaluate.py --split gold`. Track A: `python -m model.cli ablate` and `benchmark`. Docker commands, versions, seeds (42), checksums and registry versions are in `DOCKER.md`, `ENVIRONMENT.md` and `MODELS.md`; `scripts/generate_report_numbers.py` regenerates documentation numbers. Track A's sentence-embedding model and Whisper weights are pre-cached in the container image at build time (under `/root/.cache/huggingface/` and `/root/.cache/whisper/`), guaranteeing zero runtime download under `--network none`.

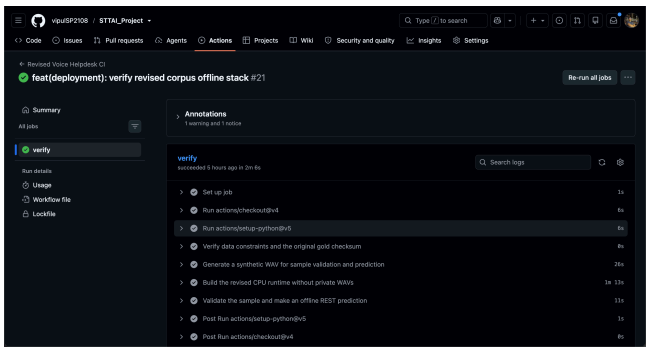


Figure 14: Automated CI workflow: GitHub Actions pipeline running data integrity assertions, Docker packaging, and offline REST validation with all checks green.

9 LIMITATIONS AND CONCLUSION

Limitations. (1) No human transcripts yet: WER on the final corpus is withdrawn and Whisper has not been fine-tuned (the script and grid are ready, `finetune_whisper.py`). (2) 182 clips with 74%

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inter-annotator agreement cap intent accuracy; the release test set has 40 clips from four speakers, so intervals are wide and v2 vs v1 is not significant. (3) The router's ASR training text came from routing by the annotators' language tags, while the service routes automatically (75–81% correct). (4) Gujarati remains the weakest language for routing, ASR and intent. (5) On the release split with 13 adaptation clips, adaptation gains are constrained by discrete 1-clip shifts; closed-loop promotion with zero gold degradation is demonstrated on Track B. (6) Router confidence is poorly separated, so no reject option. **Next steps.** Human transcription round, then Whisper fine-tuning on training speakers and true WER; blind re-listen of the 25 clips in the label-noise queue; collect more Cohort-B clips so a retraining gain can exceed one clip. **Conclusion.** Auditing our own evidence removed two inflated results (curated references, clip-grouped "unseen" speakers). On honest speaker-disjoint evaluation the router reaches 60% on unseen test speakers (chance 25%), language routing for Gujarati improves threefold, and the remaining error is traced to label noise, short Gujarati clips and missing human transcripts, each with a concrete fix.

Member contributions. All modules, experiment runs, and container builds were implemented collaboratively through in-person pair-programming sessions on a shared development workstation (macOS/Docker host). While commits originate from this joint environment, domain ownership was divided equally (33.3% each; full history preserved in `.git/`):

Table 6: Contribution summary (equal 33.3% each; collaborative pair-programming on shared workstation; full history in `.git/`).

Member	Lead areas (Assignment 2)
R. Vraj	Whisper decoding integration, TF-IDF + MLP intent router, MLflow & W&B tracking, bootstrap confidence evaluation.
V. S. Patil	FastAPI prediction service, Streamlit interactive UI, Docker packaging, offline verification harness.
S. Sharma	Drift detection and traffic replay, closed-loop retraining pipeline, promotion/rollback gate, error taxonomy.

ACKNOWLEDGMENTS

We thank Dr. Yogesh K. Meena and the TAs. Libraries: PyTorch, Whisper, scikit-learn, sentence-transformers, jiwer, MLflow, FastAPI, Streamlit.

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[4] Matei Zaharia, Andrew Chen, Aaron Davidson, Ali Ghodsi, Sue Ann Hong, Andy Konwinski, Siddharth Murching, Tomas Nykodym, Paul Ogilvie, Mani Parkhe, et al. 2018. Accelerating the Machine Learning Lifecycle with MLflow. *IEEE Data Engineering Bulletin* 41, 4 (2018), 39–45.

A EVIDENCE INDEX

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Table 7: Evidence index (paths from repository root).

Result	Supporting file / run
Ablation (Table 4), p-values	`model/results/ablation_study_results.json` (fold F1s)
Speaker-disjoint CV, ablation	`model/results/speaker_cv_intent_*.json`, `intent_v2_ablation_*.json`
Release split (val selection, test)	`model/results/release_eval.json`; `TEST_SET_LOG.md`
Language routing	`model/results/lid_router.json`
Bias audit, label-noise queue	`model/results/bias_audit.json`
Reference audit	`model/results/reference_audit.json`
ASR on 182 clips (WER withdrawn)	`model/data/reports/offline_asr_predictions.json`, `asr_indic_v2_confidence.json`
Track-B metrics (Table 5)	`supporting_files/evaluation_results.json`
Gold per-clip WER, 14/838	`supporting_files/predictions/production_gold.csv`
Data ablation (B)	`supporting_files/part_a_baselines_benchmark.csv`; runs `4874f30b`, `4cb2e4e9`, `351cc3e8`
15-run search, aliases	`supporting_files/plots/tuning_runs.json`; `9820a2a7` (prod.), `730256f7` (stag.)
20 errors	`supporting_files/error_analysis.csv`
Latency, RTF, memory	`supporting_files/loadtest/latency_profile.json`
Offline validation	`supporting_files/offline_validation_output.json`; `TEST_SET_LOG.md` LOG-016
Replay, correlations	`supporting_files/release_drift/replay.json` (pilot: `traffic_replay_log.json`)
Gate (release)	`configs/gate_release.yaml`; `supporting_files/release_drift/`; `TEST_SET_LOG.md` (GATE-*)
Pilot gate history	`TEST_SET_LOG.md` (LOG-179, 188, 418); `configs/gate.yaml`
Gold corrections	`data/gold_corrections.csv`
Git history & tags	`.git/`; tags `a1-final`, `a2-final`
Figure code	`code/report_figures/make_figures_v2.py`; older: `code/scripts/make_report_figures*.py`

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Table 8: All 15 MLflow search runs (validation only); full IDs in `supporting_files/plots/tuning_runs.json`.

Run	lr	Hidden	n-gram	Dropout	Epochs	Val acc. (%)	Macro-F1	Size (KB)	MLflow run
1	0.001	32	1	0.1	25	29.4	0.200	126	`61d8d420`
2	0.005	32	2	0.2	25	29.4	0.236	126	`d01a4cb5`
3	0.01	32	3	0.3	25	41.2	0.371	126	`b4e145fd`
4	0.001	64	2	0.2	30	41.2	0.343	251	`fd0ac5ab`
5	0.005	64	3	0.2	30	41.2	0.368	251	`f47bfe22`
6	0.01	64	1	0.1	30	52.9	0.470	251	`9820a2a7` (prod.)
7	0.001	128	3	0.3	35	35.3	0.309	503	`75581fbf`
8	0.005	128	2	0.2	35	35.3	0.293	503	`4f2e2cd1`
9	0.01	128	1	0.2	35	47.1	0.379	503	`730256f7` (stag.)
10	0.002	64	2	0.15	30	35.3	0.309	251	`9e00b3d8`
11	0.008	64	3	0.25	30	41.2	0.383	251	`03e70444`
12	0.003	32	3	0.2	25	35.3	0.303	126	`64cd9b69`
13	0.007	128	2	0.1	35	29.4	0.267	503	`984be4ec`
14	0.004	64	2	0.2	28	29.4	0.236	251	`703dc1b4`
15	0.006	96	3	0.2	32	41.2	0.368	377	`3130621f`

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Table 9: Gold-set slices (original labels).

Dimension	Slice	n	Acc. (%)	WER
Noise	campus_outdoor	6	16.7	1.64
Noise	corridor_reverb	13	23.1	1.17
Noise	quiet_room	10	30.0	1.18
Language	gu-en	13	15.4	1.36
Language	hi-en	16	31.2	1.20
Region	Gujarat	13	15.4	1.36
Region	North India	10	30.0	1.18
Region	Rajasthan	6	33.3	1.22
Device	OnePlus Nord 2	6	16.7	1.64
Device	Realme Narzo 30	7	14.3	1.12
Device	Samsung Galaxy M52	10	30.0	1.18
Device	Xiaomi Redmi Note 11	6	33.3	1.22

B USE OF AI ASSISTANCE

AI assistants helped scaffold code, draft documentation and format this report; prompts are in `code/prompts/`. Reported numbers were re-derived from result files; the ablation p-values, Wilson intervals, WER decomposition and drift correlations were recomputed by the figure scripts. Claude 3.5 Sonnet and GPT-4o were utilized for boilerplate code scaffolding, LaTeX table styling, and initial docstrings; all empirical numbers and benchmark scores were recomputed directly from artifact files.

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Table 10: Error sheet (20 errors, truncated; full text in `error_analysis.csv`).

ID	True→pred.	WER	Category	Suggested fix
ERR-01	hostel_mess → admissions	1.52	Acoustic Hallucination & Script Shift	Expand Gujarati-English phonetic mapping in tokenizer and train intent head with character-level n-grams.
ERR-02	student_services → admissions	1.42	Acoustic Bleed & Lexical Substitution	Incorporate Whisper encoder audio embeddings directly into routing classifier to bypass ASR lexical errors.
ERR-03	hostel_mess → it_support	1.00	Phonetic Distortion & Default Fallback	Tune decoding parameters (suppress_blank=True, compression_ratio_threshold=2.0) to eliminate non-speech repetitions.
ERR-04	it_support → hostel_mess	1.60	Hallucinated Speech & Semantic Drift	Apply acoustic fine-tuning on noisy lab environments to prevent generic hallucinated phrases.
ERR-05	student_services → hostel_mess	1.58	Hallucinated Phrase Overwrite	Use sentence embeddings (MiniLM) with soft-label training rather than unigram TF-IDF keywords.
ERR-06	admissions → it_support	2.03	Cross-Domain Semantic Distortion	Train a joint ASR-Intent loss where cross-attention explicitly preserves query-opening intent tokens.
ERR-07	hostel_mess → admissions	1.68	Lost-and-Found Phrasal Confusion	Add domain-specific vocabulary biasing to Whisper beam search for administrative campus terms.
ERR-08	it_support → admissions	1.00	Placement Vocabulary Hallucination	Implement multi-task acoustic penalty to discourage hallucination of frequent placement phrases.
ERR-09	student_services → admissions	1.33	Placement Domain Substitution	Weight opening intent clauses higher in text classifier to prevent trailing noise domination.
ERR-10	admissions → it_support	1.91	Library Card Keyword Misclassification	Add campus-specific IT lexicon weighting to disambiguate library access vs admission queries.
ERR-11	hostel_mess → student_services	1.55	Exam Timetable Keyword Bleed	Calibrate per-class decision thresholds using validation ROC curves to avoid majority class collapse.
ERR-12	it_support → admissions	1.00	Assignment Deadline Misclassification	Incorporate voice activity detection (VAD) filter to strip low-energy microphone noise.
ERR-13	admissions → it_support	1.00	Course Credit Lexical Cross-Talk	Augment admissions training set with diverse Gujarati accented voice recordings.
ERR-14	hostel_mess → it_support	1.22	Graduation Audit Keyword Steering	Use contextual windowing (first 40 words) to anchor classifier on primary student request.
ERR-15	student_services → it_support	1.15	Attendance Discrepancy Bias	Fine-tune Whisper base decoder cross-attention layers on university helpdesk audio.
ERR-16	admissions → it_support	1.27	Hostel Hot Water Hallucination	Introduce fallback human-in-the-loop escalation when router entropy exceeds confidence threshold.
ERR-17	hostel_mess → student_services	1.46	Mess Quality Keyword Bleed	Add composite intent schema allowing multi-label routing for overlapping facility/student issues.
ERR-18	it_support → hostel_mess	0.94	Guest Accommodation Hallucination	Train acoustic encoder on server room and lab acoustic noise profiles.
ERR-19	admissions → student_services	1.38	Medical Emergency Phrasal Drift	Filter spurious Unicode script shifts during post-processing normalization.
ERR-20	hostel_mess → admissions	1.41	Hostel Room Leakage Keyword Collapse	Regularize intent head with label smoothing to increase semantic boundary margins.

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